

Figure 2: Managing CouchDB

POST command but, additionally, pass a `-d` parameter for the data. For example:

```
# curl -X POST http://<COUCHDB_SERVER_IP>:5984/<DATABASE_NAME> \
-H 'Content-type: application/json' \
-d '{
  "fruit": "apple",
  "car": "fiat",
  "random stuff": "Some R@nD@m DATA"
}'
```

The output will contain two fields that you should make a note of:

- `"id"`: This is the unique ID that identifies your document.
- `"rev"`: This is the revision ID. Revisions are important in CouchDB, as it will not allow a user to update a document unless the document's current revision ID is provided.

```
{"ok":true,"id":"0ca55157ffdd55287d98934082b2b","rev":"1-c31f8f2f54792f63d319547db7180"}
```

You can list any document's contents by using the `curl GET` command along with the document's ID as shown:

```
# curl -X GET http://<COUCHDB_SERVER_IP>:5984/<DATABASE_NAME>/<DOCUMENT_ID>
```

Deleting a database is also a simple process; just provide the database's name with the `curl DELETE` parameter as shown below:

```
# curl -X DELETE http://<COUCHDB_SERVER_IP>:5984/<DATABASE_NAME>
```

By now you may be thinking that you need to be a command line Ninja to manage CouchDB. But that's not true because there are Web based tools, like Futon, to manage CouchDB. Futon is a Web based user interface provided by CouchDB that helps in the creation, updating

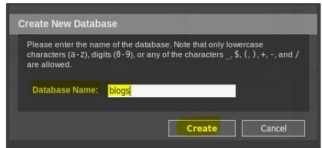


Figure 3: Creating a database using Futon

and overall management of the databases. You can additionally create and work with documents, list various database configuration parameters and initiate database replication as well, using this UI.

To access Futon, simply launch a browser and type the following command:

```
http://<COUCHDB_SERVER_IP>:5984/_utils
```

You will be presented with a dashboard that contains a 'Home' screen or the 'Overview' tab. This tab shows you a list of databases that you have created on the current CouchDB instance. To the right of the UI, there is a 'Tools' section, where you can select different options that will guide you through changing the database's configuration, enable replication, etc. You can also view the current status and health of your CouchDB instance using the 'Tools' section.

To create a new database, select the 'Create Database' button. Provide a suitable name for your database and click 'Create' once you are done.

Now that you have created your database, you can populate it with documents, manage the database's security and even delete it, if required.

Replication

The replication process in CouchDB is an incremental one-way process involving two or more CouchDB database instances. These instances can be on the same CouchDB server or running on a separate node altogether.

The main purpose of replication in CouchDB is that, at the end of the process, all active documents from the source database are successfully placed on the remote database as well. Alternatively, if the database is deleted from the sources' end, then the same should also get deleted from the remote database as well.

In this case, we have two CouchDB instances on two different nodes (DB-HOST-1 and DB-HOST-2). Each CouchDB instance has an empty database created within it. On DB-HOST-1, we have a database called `db-01` and on the second host, we have a database called `db-02`.

To start the replication process, we make use of the `_replicate` database/interface. Here, you need to provide the source and target database information as shown below: